

RAG vs. LoRA Fine-Tuning for Course-Specific Question Answering

Tamar Telele | Applied Language Models | Dr. Barak Or | Google & Reichman University

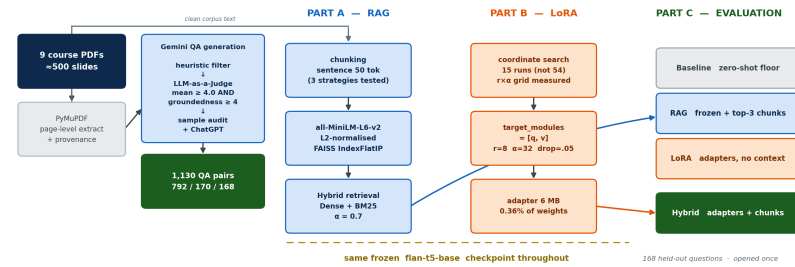


Figure 1 — System overview. Nine course PDFs become 1,130 audited QA pairs, which feed two independently-tuned adaptation paths. The dashed rule is the methodological core of this work: Baseline, RAG, LoRA and Hybrid all run on **one frozen flan-t5-base checkpoint**, so every measured difference is attributable to method rather than to model capacity. The 168-question test set is opened exactly once, after every hyperparameter has been fixed on validation.

Abstract — RAG and parameter-efficient fine-tuning are usually presented as competing strategies for adapting a pretrained model to a narrow domain. We test that framing directly. A 1,130-pair QA dataset was constructed from nine lecture-slide PDFs through a three-stage generation-and-audit pipeline, and three systems were built on the same frozen **flan-t5-base** checkpoint: RAG (frozen weights + retrieval), LoRA (adapters, no retrieval) and Hybrid (adapters + retrieval). Hybrid wins every metric (ROUGE-L 0.444, BLEU 0.238) — a 2.6× ROUGE-L and 15.9× BLEU gain over the zero-shot floor — while matching RAG's hallucination rate. Three independent measurements locate the bottleneck in the reader rather than the index, and an NLI-based analysis establishes that fabrication and low lexical overlap are statistically independent failure modes.

1 · Introduction

Adapting a general model to a specific body of knowledge admits two architecturally distinct answers. **RAG** leaves the weights untouched and injects evidence into the prompt at inference; knowledge lives outside the model and updates by re-indexing. **LoRA** writes knowledge into the weights through low-rank adapters; nothing is retrieved, but knowledge freezes at training time. Opposite update costs, opposite latency profiles and — as this work shows — opposite failure modes.

Published comparisons typically differ in base model, corpus and evaluation protocol simultaneously, which makes the causal attribution ambiguous. This project removes those confounds: one corpus, one base checkpoint, one test set opened exactly once. The question is stated precisely — *when does retrieval beat parameter-efficient fine-tuning on course-specific QA, and why?*

A third condition, **Hybrid**, converts the question into a falsifiable prediction. If the two mechanisms capture the same signal, Hybrid should match the better of the two. If they capture different signal, Hybrid should exceed both. §6 reports which happened.

2 · Dataset Construction

No public dataset exists for this corpus, so it was built end-to-end. Nine course PDFs were extracted page by page with PyMuPDF, preserving *file + page* provenance for every span. In lecture slides a page approximates one idea, which makes it a natural extraction unit and — critically — gives every QA pair an automatic ground-truth source with **no manual labelling**, which is what makes the retrieval evaluation in §3.4 and the error taxonomy in §7 possible at all.

2.1 Cleaning and generation

A nine-step deterministic pipeline was applied: lowercasing, whitespace normalisation, removal of short lines, decoration symbols, slide numbers, URLs and duplicate sentences. Gemini then generated QA pairs from overlapping windows that never cross document boundaries. Three prompt decisions determined quality: a **strict grounding instruction**, **structured JSON output** rather than regex-scraped prose, and an **explicit question-type mix**. The last is not cosmetic — a dataset of only "what is X?" cannot discriminate between RAG and LoRA, since both handle definitions well; divergence appears on conceptual and comparative items, so those had to be present by design.

2.2 Three-stage quality protocol

Stage	Criterion
1 — Heuristic	Length bounds; degenerate answers ("yes", "n/a"); leakage phrases ("according to the slide"); exact duplicates; near-duplicates at Jaccard ≥ 0.85
2 — LLM Judge	Gemini at temperature 0.0 scores four criteria 1–5: groundedness, correctness, clarity, self-containment. Accept iff mean ≥ 4.0 and groundedness ≥ 4
3 — Human + GPT	Seeded random sample audited manually alongside the machine scores; ChatGPT produced separate deletion and correction lists, applied programmatically

The judge is the generator, which sounds circular — so the setup is **asymmetric by design**. The judge sees the *source context* and answers a verification question ("is this

answer supported by this text?"), a strictly easier and better-calibrated task than open generation, scoring four independent criteria at temperature 0.0.

The acceptance rule deserves emphasis: **mean ≥ 4.0 and groundedness ≥ 4 , evaluated separately**. A pair can average 4.2 on the strength of clarity and self-containment while scoring 3 on groundedness — fluent, well-formed, and not actually supported by the slides. Such a pair would corrupt the RAG index and poison LoRA's training signal simultaneously. The conjunctive rule makes that failure mode unreachable.

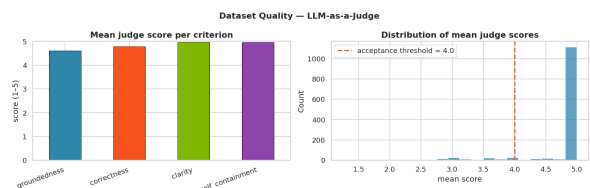


Figure 2 — Judge calibration. All four criteria average above 4.6. Groundedness scores lowest by design: it is the strictest constraint, demanding traceability to a specific slide rather than to general knowledge. The distribution is healthy — dense mass at 5.0 with a small tail of legitimate rejects, which is what a working filter looks like. A filter that rejected nothing would indicate a judge that is not discriminating. 1,140 of 1,279 pairs passed (89.1%).

The manual audit estimated the **judge's own error rate** rather than fixing individual cases. A seeded random sample — reproducible, immune to the selection bias of hand-picking — was displayed inline with the machine scores and exported as an audit sheet with an empty verdict column, against a criterion fixed in advance at $\geq 90\%$ correct. After ChatGPT-assisted deletions and corrections, **1,130 pairs** remained.

2.3 Composition and split

The mix is **54.4% factual / 34.8% conceptual / 10.8% comparative**; questions average 11 words and answers 17 — deliberately short and grounded, with no multi-paragraph answers that would reward verbosity over accuracy under n-gram metrics. All nine lectures contribute, with Classical ML leading at 221 pairs.

The split is 70/15/15 — **792 / 170 / 168** — stratified by **source PDF rather than topic**. This corrects the obvious choice: the LLM assigns free-text topic labels, which produced over 500 distinct values, so stratifying on them would have been random assignment wearing a stratification label. Source PDF is a small, clean, semantically meaningful partition. Deduplication runs *before* the split to prevent leakage, and the test set was written once and untouched until §6.

3 · Part A — RAG Pipeline

3.1 Chunking: a measured ablation

Chunk size is the first irreversible decision in a RAG system. Too small and the chunk loses the context that makes it interpretable; too large and the embedding is diluted across several ideas and precision collapses. Three granularities were therefore measured on validation MRR: **sentence** (50 tok / 10 overlap), **paragraph** (300/50) and **slide** (500/75).

Two implementation details carry the rest of the project. The splitter is constructed **from the flan-t5 tokenizer**, so 50/300/500 are true token counts rather than a character heuristic that would drift silently per document; and **add_start_index=True** maps every chunk back to its source PDF and page range. This is a *granularity ablation* — the splitter and separator hierarchy are fixed so exactly one variable moves, which is what licenses a causal reading. The strategies produce an **11× spread in chunk count** (1,407 sentence vs 125 slide), and realised size tracks the target except at sentence level, where the 39-token average falls short of the 50-token budget — sentence boundaries, not the token limit, bind at that scale.

3.2 The embedding space

Chunks are encoded with **all-MiniLM-L6-v2** (384 dimensions, 22M parameters) and **L2-normalised**, which turns the inner product into exact cosine similarity. That is not a stylistic choice: a bounded $[-1, 1]$ score that is comparable across questions is a precondition for the single global threshold calibrated in §3.5. The index is therefore **FAISS IndexFlatIP** rather than IndexFlatL2 — mathematically equivalent after normalisation, but IP returns the similarity the guardrail consumes directly. *Flat* means exact exhaustive search: milliseconds at this scale, and zero approximation error contaminating the experiment.

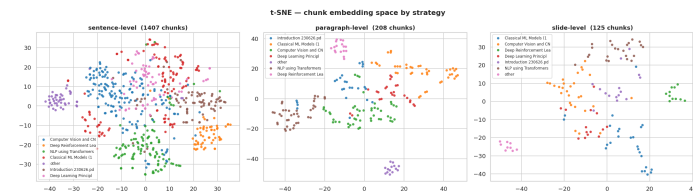


Figure 4 — t-SNE of chunk embeddings, all three granularities, coloured by source PDF. The learned structure is semantic rather than lexical: topics that share core vocabulary sit close together — Classical ML, the Introduction lecture and Deep Learning Principles overlap substantially, reflecting shared foundations such as gradient descent and loss functions — while Deep Reinforcement Learning, whose vocabulary (reward, policy, Bellman) barely overlaps with the rest of the curriculum, separates into its own isolated region on every granularity. **Cluster separation is visibly sharpest at paragraph level (centre)**, where boundaries are distinct; sentence-level (left) is diffuse because short fragments carry too little context for the encoder to anchor them to one topic, and slide-level (right) clusters cleanly but at low density. The natural inference is that paragraph-level should retrieve best. **It does not — sentence-level wins every configuration in §3.4, and that discrepancy is itself the finding.**

3.3 Three retrievers, three failure modes

Dense ranks by cosine similarity in the shared 384-dimensional space. Its strength is meaning: a question about reducing parameter count retrieves a LoRA chunk with near-zero vocabulary overlap. Its weakness is that technical tokens — **ROUGE-L**, **flan-t5-base** — are averaged into one vector and lose discriminative power. **BM25** is the targeted complement: Okapi scoring excels at precisely those acronyms and model names, and fails on vocabulary mismatch.

Hybrid fuses them as $\alpha \tilde{s}_{\text{dense}} + (1 - \alpha) \tilde{s}_{\text{BM25}}$, where $\tilde{s}$ is min-max normalisation **per query, not per corpus** — a load-bearing detail, since BM25 scores are unbounded and corpus-dependent, so without it BM25 would silently dominate the sum and α would not mean what it appears to mean.

One diagnostic question makes the case concrete. For *"What are two examples of supervised learning provided in the course material?"* Dense returned 2 relevant chunks of 3, BM25 returned 1, Hybrid returned 2 with more concentrated scores. Dense won because the question is conceptual; BM25 failed because *learning* is shared with unrelated chunks. The point is not that Dense is better — it is that **no single method wins consistently**, which is exactly the argument for fusion.

3.4 Grid search and the sentence-level result

Relevance ground truth comes free from provenance: a chunk is relevant if its source file matches and its page range overlaps within ± 1 page. **MRR was selected as the optimisation target over Hit@3**, and the reasoning matters. Each pair has exactly one ground-truth source. A retriever that surfaces it at rank 3 still scores $\text{Hit}@3 = 1$ — but the reader then receives two irrelevant chunks *before* the correct one, and that context pollution degrades generation. MRR penalises it directly, contributing 0.33 at rank 3. Optimising MRR aligns retrieval quality with end-to-end answer quality, which is the quantity that actually matters.

Configuration	MRR	Hit@1	Hit@3
sentence + Hybrid	0.706	0.565	0.847
sentence + Dense	0.687	0.547	0.812
sentence + BM25	0.657	0.529	0.771
paragraph + Hybrid	0.670	0.506	0.835
paragraph + Dense	0.617	0.459	0.741
slide + Hybrid	0.620	0.494	0.724

Two clean results. **Hybrid wins on all three chunking strategies**, confirming that fusion contributes real value rather than only complexity. And **sentence-level wins on all three retrieval methods**, contradicting the visual intuition from Figure 4.

The reconciliation is that the two measure different properties. t-SNE asks whether a chunk is *topically identifiable*, and a 300-token paragraph obviously is. Retrieval asks whether a chunk *answers a specific question*, and there a 50-token window holding one fact with no surrounding noise is a far better match. Larger chunks dilute the relevant signal with context that is thematically related but not answer-bearing. **Granularity must be measured against the actual objective, not inferred from cluster quality.**

An α sweep on sentence-level Hybrid peaks at $\alpha = 0.7$ (MRR 0.717, Hit@1 0.576, Hit@3 0.841) — 70% Dense, 30% BM25. The curve is smooth and single-peaked, so the optimum is a property of the data rather than a fortunate grid point. The value is interpretable. The retriever leans on dense embeddings with BM25 acting as a lightweight corrector, which is the right shape for this corpus: lecture language is internally consistent, so semantic embeddings generalise well across paraphrase, and BM25 earns its 30% mainly when a question reuses exact slide vocabulary or hinges on a technical token that dense averaging would wash out.

3.5 The abstention guardrail

A study assistant that confidently answers an off-syllabus question is worse than useless, so the system abstains when maximum similarity falls below a threshold. That threshold was obtained by **calibration**: top-1 similarity was computed for every in-scope validation question and for 12 deliberately out-of-scope probes spanning cooking, football and astronomy, then candidate thresholds were swept and the value maximising **balanced accuracy** was retained. Balanced accuracy is the correct objective here because both error directions carry real cost and the two classes are artificially balanced — optimising raw accuracy would simply let the larger class dominate. Threshold **0.330** achieves **100% balanced accuracy**: the 5th percentile of

in-scope similarity (0.462) sits well above the 95th percentile of out-of-scope similarity (0.294), a 0.168 margin with essentially no overlap, and accuracy peaks across a wide band (≈ 0.30 – 0.35) rather than at a single fragile point.

3.6 Reader and prompt selection

The retriever finds evidence; the reader turns it into an answer. The reader is the **untouched flan-t5-base** — the same checkpoint Part B fine-tunes, which is what keeps the Part C comparison clean. Decoding uses beam search (**num_beams=4**) with **do_sample=False** so output is reproducible, and **min_new_tokens=15** to eliminate degenerate single-token outputs that occur when EOS is the locally optimal first token.

Prompt wording changes how effectively a reader exploits retrieved evidence, so three variants were compared on validation and **selected by ROUGE-L — the same metric used throughout every other selection decision in this project**, so that no stage is optimised against a different objective than the rest. **v1** (role framing with task-first ordering) won over **v2** (explicit material delimiters plus a length constraint) and **v3** (added paraphrase guidance). Direct chunk context also outperformed Small-to-Big parent-document expansion: at 39 tokens per chunk the top-3 context already fits comfortably inside the 512-token encoder budget, so expansion adds noise without adding headroom.

4 • Part B — LoRA Fine-Tuning

Adapters are applied to the query and value projections only (**target_modules=["q", "v"]**). Hu et al. (2021) showed that adapting Q and V alone matches full attention fine-tuning, while adding K and O costs parameters for negligible gain. Fixing this also keeps the coordinate search clean — one factor moves at a time. A single entry point runs every experiment, so configuration, metrics, training curve and runtime land in one comparable table.

4.1 Coordinate search: 15 runs instead of 54

A full grid over 3 tokenizations $\times$ 3 ranks $\times$ 3 alphas $\times$ 2 dropouts is 54 runs. Coordinate search fixes one factor at a time at the best value so far, costing 11 — but it is blind to interactions, and there is exactly one place an interaction is genuinely expected. The adapter contributes (α/r) -BA, so α and r are **not independent**; their ratio is what matters, which is where the $\alpha = 2r$ heuristic originates. Four grid-completion runs were added so **every cell of the $r \times \alpha$ heatmap is measured rather than interpolated**. Total: **15 runs, a 3.6 $\times$ reduction**.

Stage	Varies	Selected on	Runs
A	word / subword / char	val ROUGE-L	3
B	$r \in \{4, 8, 16\}$	val ROUGE-L	3
C	$\alpha \in \{8, 16, 32\}$ + grid	val ROUGE-L	3 + 4
D	dropout $\in \{0.05, 0.1\}$	val loss	2

4.2 Stage A — tokenization as a controlled baseline

Quality and cost were compared side by side, because a scheme that wins by 0.002 ROUGE-L while tripling sequence length is not a win but a bad trade. Word and subword segmentation are **statistically indistinguishable** on ROUGE-L (0.2530 vs 0.2521); the T5 vocabulary handles both well and the 0.0009 gap sits inside validation noise. Word was selected. Character-level was included as a **controlled baseline rather than a serious candidate**: it expands average input length 6 $\times$ while halving ROUGE-L to 0.129. The value is in the magnitude, not the direction — the 50% degradation quantifies the cost of losing term identity, which is the *upper bound* on what tokenization alone can affect in this setting. Strings like **LoRA** or **ROUGE-L** do not survive character-level fragmentation.

4.3 Capacity, scaling and regularisation

r is the rank of the low-rank update, i.e. adapter capacity. Too small and the adapter cannot represent the task; too large and it memorises a 792-example training set while parameter count grows linearly. **r = 8 won**, and the failure of $r = 16$ is informative: despite twice the parameters it scored *worse*, with higher eval loss (2.60 vs 2.54) — the signature of memorisation rather than generalisation. The bias–variance trade-off appears directly in a hyperparameter sweep.

$\alpha = 32$ won, giving $\alpha/r = 4.0$, double the conventional heuristic. On a small dataset a higher scaling factor accelerates learning before overfitting sets in, and the learning curve confirms it. **Dropout was selected on validation loss rather than ROUGE-L**, deliberately: loss is the more sensitive over-fitting signal, and by Stage D the ROUGE gap between dropout values is inside the noise. Using a noisy metric to make a fine-grained decision is how spurious hyperparameters get chosen.

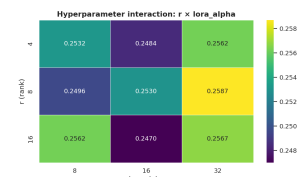


Figure 8 — $r \times \alpha$ heatmap, every cell measured. The completed grid confirms the coordinate search: ($r = 8, \alpha = 32$) is the true optimum, validating the coordinate assumption on this dataset rather than merely asserting it. The instructive exception is that the $\alpha = 16$ column is uniformly weak regardless of rank, indicating scaling that is genuinely too low rather than a rank-specific effect.

Final configuration: word tokenization, r = 8, α = 32, dropout = 0.05 — 884,736 trainable parameters out of 248M (0.36%).

4.4 Final training

The winning configuration was retrained against the full validation set (the search used a 100-example subset for speed), reaching ROUGE-L 0.2612 — the highest of all 16 runs, confirming it is real rather than an artefact of the subset. Only adapter weights are saved: 6 MB against a 990 MB base — the practical argument for LoRA in production. The curves show no train/validation divergence, both metrics plateau at epoch 2, and ROUGE-L and BLEU track each other throughout, so the choice of selection metric does not change the outcome.

5 · Evaluation Protocol

This is the first and only use of the test set. Every decision up to this point — chunk size, retrieval method, α, guardrail threshold, prompt variant, tokenization, r, α, dropout — was made on validation. Four systems ran on the same 168 questions: Baseline (frozen base, no context), RAG (frozen base + top-3 chunks), LoRA (adapters, no context) and Hybrid (adapters + top-3 chunks).

The Baseline was included deliberately as a zero-shot floor. Without it a ROUGE-L of 0.44 is a number without a scale; with it, the gap from 0.168 to 0.444 is unambiguously attributable to method, since the underlying weights are identical across all four conditions.

ROUGE-L is the selection metric throughout — for chunking, retrieval, prompt choice and every LoRA stage — and BLEU is reported alongside it as an independent check. Using one metric consistently across all selection decisions avoids the subtle failure where different pipeline stages are optimised against different objectives. BLEU is the stricter of the two, since n-gram precision penalises omitting or substituting precise technical terms, so agreement between them is informative: the two produce identical rankings, so the conclusion does not depend on metric choice. Latency is reported as a deployment consideration — RAG pays at inference, LoRA once at training time.

6 · Results

System	ROUGE-L	ROUGE-1	BLEU	Latency
Baseline	0.1679	0.2071	0.0150	0.67 s
RAG	0.3894	0.4328	0.2036	0.75 s
LoRA	0.2728	0.3247	0.0569	1.10 s
Hybrid	0.4436	0.4805	0.2381	1.05 s

Hybrid wins every quality metric — 2.6× ROUGE-L and 15.9× BLEU over the baseline, +0.054 ROUGE-L over RAG alone and +0.171 over LoRA alone. The prediction from §1 is confirmed: Hybrid exceeds both components rather than matching the better of them, which is only possible if retrieval and fine-tuning contribute different signal.

The ordering RAG > LoRA is equally informative. With only 792 training examples, adapters cannot absorb the factual content of nine lectures — but retrieval can surface any of it on demand. LoRA's contribution is therefore not knowledge but fluency and answer shape, which is precisely what Hybrid exploits. A five-dimensional normalised comparison confirms the shapes differ in kind, not only size: RAG leads on speed and coverage, LoRA on conceptual questions, and Hybrid is outermost on every quality axis at once.

6.1 Performance by question type

Type	RAG	LoRA	Hybrid	Δ vs RAG
Factual	0.3975	0.2886	0.4500	+0.053
Conceptual	0.4049	0.2551	0.4395	+0.035
Comparative	0.3187	0.2655	0.4328	+0.114

Comparative questions are where the architecture earns its keep. RAG is weakest there (0.319) — comparison requires synthesising several facts, and a frozen reader handed three chunks tends to echo one. Hybrid's +0.114 is more than double its gain on any other type: retrieval supplies the facts to compare, the adapter supplies the learned pattern of how a comparison is phrased. This is the clearest single piece of evidence for complementarity in the study.

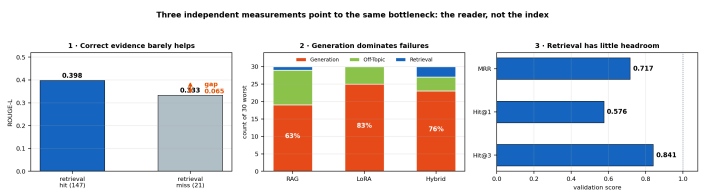


Figure 11 — Three independent measurements locate the same bottleneck. (1) Splitting RAG's test answers by retrieval outcome gives ROUGE-L 0.398 on hits versus 0.333 on misses — a gap of only 0.065 at an 87.5% hit rate. If retrieval quality were the binding constraint this gap would be large, because finding the correct chunk would transform the answer. It does not. (2) Classifying the 30 worst answers per system shows generation error dominating at 63–83% while retrieval error is nearly absent — the same conclusion from an entirely different measurement. (3) Validation retrieval scores are already high (MRR 0.717, Hit@3 0.841), leaving little headroom. Together these invert the intuitive engineering priority: effort on embeddings or chunking returns little; effort on the generation stage returns a great deal.

7 · Error Analysis and Hallucination

The 30 worst answers per system were classified using signals already collected — retrieval hit/miss, lexical overlap, guardrail score, context length — so the taxonomy is reproducible rather than impressionistic. Off-Topic = essentially no lexical overlap with the reference (ROUGE-L < 0.05), i.e. a different question was answered; Generation Error = on-topic but incorrect or incomplete; Retrieval Error = the correct chunk never entered the top-k.

Category	Hybrid	LoRA	RAG
Generation Error	23	25	19
Off-Topic	4	5	10
Retrieval Error	3	0	1
Knowledge Gap	0	0	0
Truncation Error	0	0	0

Zero Truncation Errors validates the sentence-level choice: at 39 tokens per chunk, three chunks never approach the 512-token limit. Zero Knowledge Gaps means the guardrail never fired spuriously in-scope. RAG's Off-Topic count (10) is double Hybrid's (4) — the adapter's contribution made visible: given a marginally relevant chunk the frozen reader echoes whatever it discusses, while the fine-tuned reader stays anchored to the answer shape.

7.1 NLI-based hallucination detection

Hallucination is fabrication: asserting a specific fact that is not true. No lexical statistic can detect it, because fabrication is a semantic property — a fabricated answer can reuse the reference's vocabulary almost entirely while inverting its meaning. We therefore framed the problem as natural language inference: treat the reference answer as premise and each claim in the prediction as hypothesis, and ask a DeBERTa-v3 NLI cross-encoder whether the claim is entailed, neutral, or contradicted.

Each prediction is split into claims at sentence level, discarding fragments under five words, and flagged only if at least one claim is contradicted with confidence > 0.70. Entailment and neutral are never penalised, so a correct paraphrase and a vague-but-harmless answer both pass unflagged. Only assertions that conflict with the reference count — which is what separates hallucinating from being merely wrong.

System	Halluc. rate	Flagged	ROUGE-L
RAG	11.9%	20 / 168	0.389
LoRA	14.3%	24 / 168	0.273
Hybrid	11.9%	20 / 168	0.444

LoRA hallucinates most, exactly as the architecture predicts: with no retrieved evidence it answers from parametric memory and invents confidently, attributing TensorFlow to UC Berkeley and PyTorch to Carnegie Mellon — both fabrications, both fluent, both at contradiction 1.000. RAG and Hybrid are identical at 11.9%, localising the effect precisely: retrieval suppresses fabrication, and the adapter neither helps nor hurts on this axis. Hybrid therefore buys accuracy without paying in hallucination.

7.2 High confidence, wrong answer

Live probing surfaced a failure the aggregate metrics conceal. Asked "What is the function of convolutional layers in a CNN?" — where the reference is "They extract features from the input image" — the retriever returned chunks at similarity 0.997, 0.964 and 0.896, all from the correct PDF and all about CNNs. Every system still answered incorrectly. RAG produced "visualizing which parts of an image activate the neurons in a convolutional neural network"; Hybrid produced "the first convolutional layer specifies the input shape (28×28×1) and uses 32 filters of size 3×3". Both are statements about convolutional layers. Neither answers the question.

A near-perfect similarity score is not evidence of an answer-bearing chunk. Cosine similarity measures topical proximity; it cannot distinguish a chunk that is about convolutional layers from one that answers a question about them. This is the mechanism behind the Off-Topic category, invisible to any retrieval metric scoring relevance by provenance — including the Hit@k and MRR used here, which would both record this as a successful retrieval. It is the strongest argument in this work for cross-encoder reranking.

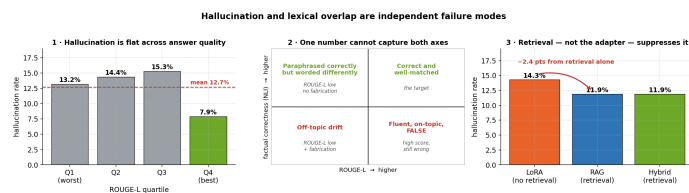


Figure 12 — Hallucination and lexical overlap are independent failure modes. (1) Hallucination rate by ROUGE-L quartile is essentially flat (13.2 / 14.4 / 15.3%) across Q1–Q3, dropping only in the best quartile (7.9%). If low ROUGE-L and fabrication were the same phenomenon, the rate would fall monotonically. (2) The two axes define four failure regions and only one is the target: an answer can score low because it paraphrases correctly but differently (upper-left, no factual problem), or score high while asserting something false (lower-right, fluent and wrong). A single quality number collapses these and misattributes the root cause. (3) Isolating the mechanism: LoRA without retrieval hallucinates at 14.3%; adding retrieval brings RAG and Hybrid to 11.9% — a 2.4-point reduction attributable to retrieval alone.

8 • Limitations

- **Reference-answer noise sets the ceiling.** Answers were LLM-generated and verified on an audited sample. The ranking across systems is trustworthy; the absolute ceiling reflects the reference set, not the systems.
- **Generation capacity is the binding constraint.** flan-t5-base has 250M parameters; the 0.065 hit/miss gap and 63–83% generation-error share both show retrieval has little headroom left.
- **Granularity, not algorithm.** The chunking ablation varies size with the splitter fixed; semantic and page-boundary chunking were not tested, and coordinate search measured only the $r \times \alpha$ interaction.
- **NLI inherits its model's calibration.** A fabricated fact *absent from* rather than *conflicting with* the reference registers as neutral, so rates are a **lower bound**. Each config was trained on one seed; gaps under ~ 0.01 ROUGE-L are not significant.

9 • Conclusion and Future Work

Retrieval and parameter-efficient fine-tuning are **complementary mechanisms, not competing ones**. Hybrid exceeds both components on every metric, which is only possible because they contribute different signal: retrieval supplies facts the adapter never had capacity to absorb, and the adapter supplies answer structure the frozen reader never learned. The hallucination analysis localises each contribution precisely — retrieval suppresses fabrication (14.3% $\rightarrow$ 11.9%), while the adapter improves quality without affecting factuality in either direction.

The strongest practical finding is that **the reader, not the retriever, is the ceiling**, supported by three independent measurements. This inverts the intuitive engineering priority and determines what is worth doing next.

- **Replace the reader.** The generation stage is the measured bottleneck, so substituting a larger base model — a quantised 7–8B reader on the identical retrieval pipeline — is the single change with the highest expected return, and would also quantify exactly how much of the ceiling is model capacity.
- **Replace the QA-generation model.** The reference answers, and therefore the metric ceiling, are products of a single generator. Regenerating the corpus with a stronger model — or an ensemble keeping only pairs several generators agree on — would raise ground-truth quality and reduce the residual noise identified in §8. This attacks the evaluation ceiling rather than the system, and both need to move.
- **Hybrid fine-tuning.** Train the adapter *with retrieved context inside the prompt*, so it learns to read evidence rather than to recall facts, attacking the measured bottleneck within the current model size.
- **Cross-encoder reranking.** Rerank the top-k by query–chunk interaction rather than embedding distance, addressing the 0.997-similarity failure in §7.2 that no bi-encoder can detect by construction.
- **Question-type routing and semantic evaluation.** A lightweight classifier routing factual $\rightarrow$ RAG, conceptual $\rightarrow$ LoRA, comparative $\rightarrow$ Hybrid could outperform any fixed architecture; and BERTScore or a calibrated LLM judge as the primary metric would remove the reference-phrasing ceiling entirely.